Lesson 8: Probability density functions (PDFs)

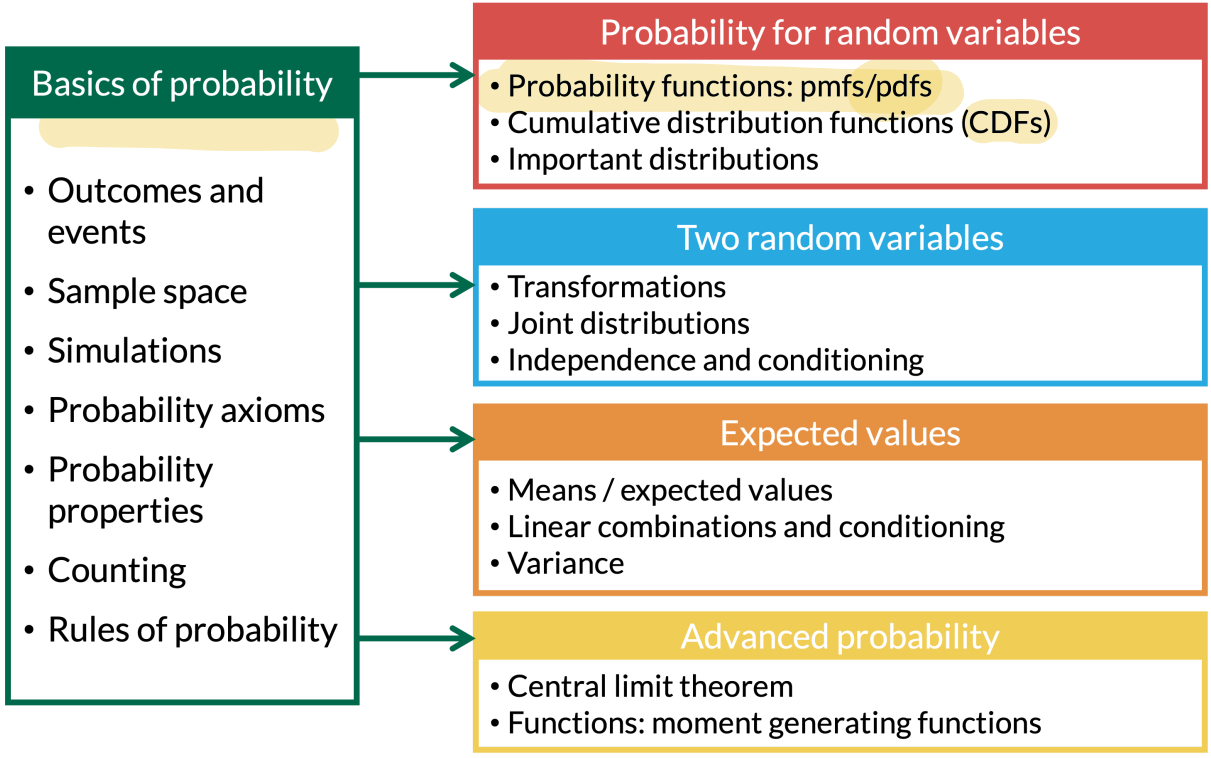
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Learning Objectives

1. Distinguish between discrete and continuous random variables.
2. Calculate probabilities from pdfs for continuous random variables.
3. Use R to simulate known continuous distributions.

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

Learning Objectives

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Discrete vs. Continuous RVs

- For a **discrete** RV, the set of possible values is either finite or can be put into a countably infinite list. $P(X = x)$
- **Continuous** RVs take on values from continuous *intervals*, or unions of continuous intervals

DISC RV $\rightarrow$ $p_X(x) = P(X = x)$

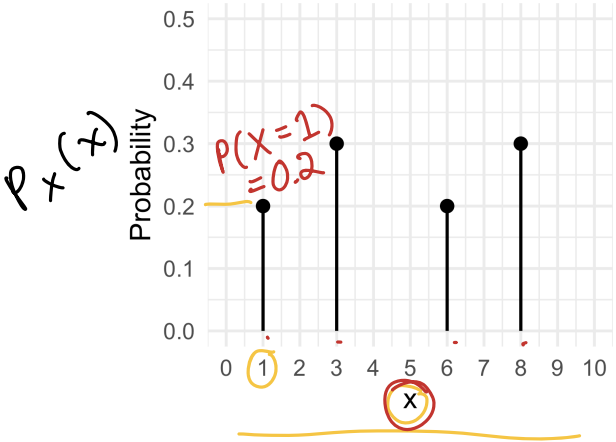
cont RV $\rightarrow$ $f_X(x) \neq P(X = x)$

	Discrete	Continuous
probability function	mass (probability mass function; PMF)	density (probability density function; PDF)
	$0 \leq p_X(x) \leq 1$	$0 \leq f_X(x)$ (not necessarily ≤ 1)
	$\sum_x p_X(x) = 1$	$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$
	$P(0 \leq X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$ if X is integer valued	$P(0 \leq X \leq 2) = \int_0^2 f_X(x)\,dx$
	$P(X \leq 3) \neq P(X < 3)$ when $P(X = 3) \neq 0$	$P(X \leq 3) = P(X < 3)$ since $P(X = 3) = 0$ always
cumulative distribution function (CDF) $F_X(x)$	$F_X(a) = P(X \leq a) = \sum_{x \leq a} P(X = a)$ graph of CDF is a step function with jumps of the same size as the mass, from 0 to 1	$F_X(a) = P(X \leq a) = \int_{-\infty}^{a} f_X(x)\,dx$ graph of CDF is nonnegative and continuous, rising up from 0 to 1
examples	counting: defects, hits, die values, coin heads/tails, people, card arrangements, trials until success, etc.	lifetimes, waiting times, height, weight, length, proportions, areas, volumes, physical quantities, etc.
named distributions	Bernoulli, Binomial, Geometric, Negative Binomial, Poisson, Hypergeometric, Discrete Uniform	Continuous Uniform, Exponential, Gamma, Beta, Normal
expected value	$\mathbb{E}(X) = \sum_x x p_X(x)$ $\mathbb{E}(g(X)) = \sum_x g(x) p_X(x)$	$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx$ $\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$
$\mathbb{E}(X^2)$	$\mathbb{E}(X^2) = \sum_x x^2 p_X(x)$	$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx$
variance	$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$	$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$
std. dev.	$\sigma_X = \sqrt{\mathrm{Var}(X)}$	$\sigma_X = \sqrt{\mathrm{Var}(X)}$

Figure from Introduction to Probability TB (pg. 301)

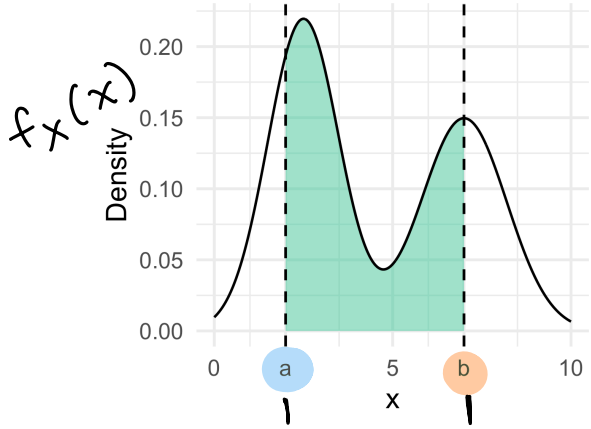
How to define probabilities for continuous RVs?

Discrete RV X:



- pmf: $p_X(x) = P(X = x)$

Continuous RV X:



- density: $f_X(x)$
- probability: $P(a \leq X \leq b) = \int_a^b f_X(x)dx$

↳ area under the curve from a to b

What is a probability density function?

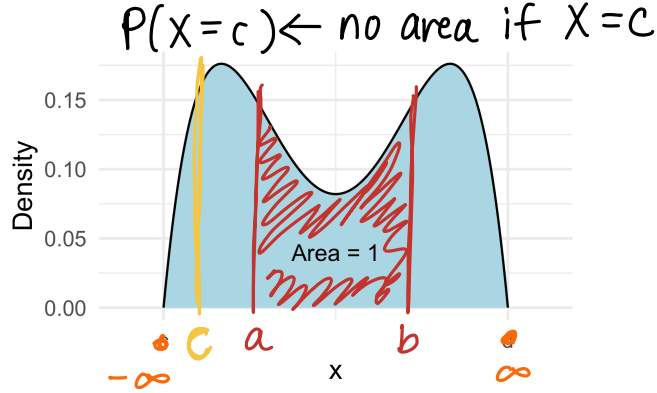
Probability density function

The probability distribution, or **probability density function (pdf)**, of a continuous random variable X is a function $f_X(x)$, such that for all real values a, b with $a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)dx$$

Remarks:

1. Note that $f_X(x) \neq \mathbb{P}(X = x)$!!!
2. In order for $f_X(x)$ to be a pdf, it needs to satisfy the properties
   - $f_X(x) \geq 0$ for all x
   - $\int_{-\infty}^{\infty} f_X(x)dx = 1$: the total area under the curve is 1



Learning Objectives

1. Distinguish between discrete and continuous random variables.

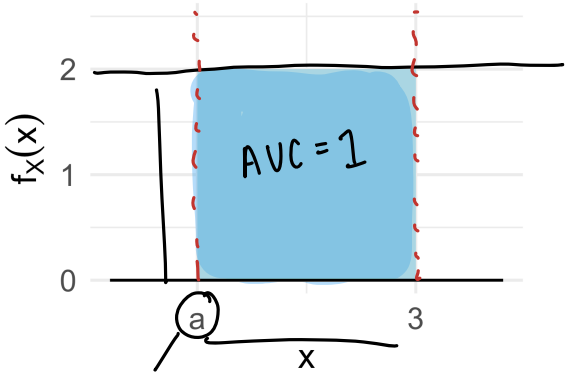
2. Calculate probabilities from pdfs for continuous random variables.

3. Use R to simulate known continuous distributions.

Example 1.1

Let $f_X(x) = 2$, for $a \leq x \leq 3$.

1. Find the value of a so that $f_X(x)$ is a valid pdf.



$= 2.5$ $(3-a) \times 2 = 1$

$0.5 \times 2 = 1$ ✓

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

$$\int_{a}^{3} 2\,dx = 1$$

$$2x \Big|_{x=a}^{x=3} = 1$$

$$2(3) - 2a = 1$$

$$6 - 2a = 1$$

$$5 = 2a$$

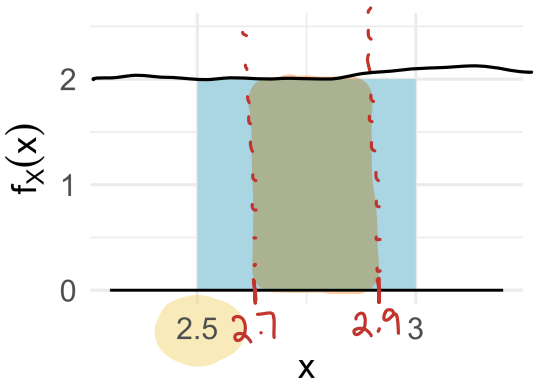
$$a = 5/2 = 2.5$$

Example 1.2

Let $f_X(x) = 2$, for $2.5 \leq x \leq 3$.

2. Find $\mathbb{P}(2.7 \leq X \leq 2.9)$.

↳ bounds of pdf



$2 \cdot (2.9 - 2.7) = 0.4$

$$P(2.7 \leq X \leq 2.9) = \int_{2.7}^{2.9} 2 \, dx$$

2.9 — make sure integrands w/in bounds

2.7

↳ intergrands

$$= 2x \Big|_{x=2.7}^{x=2.9}$$

$$= 2(2.9) - 2(2.7)$$

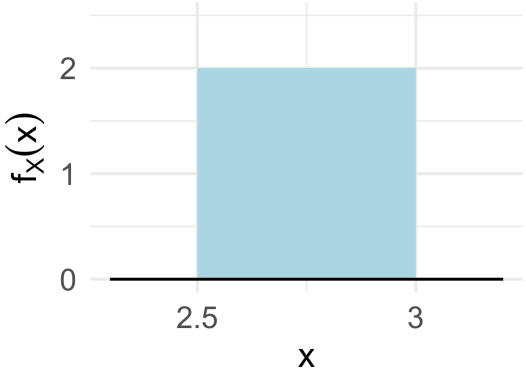
$$= 0.4$$

$$P(2.7 \leq X \leq 2.9) = 0.4$$

Example 1.3

Let $f_X(x) = 2$, for $2.5 \leq x \leq 3$.

3. Find $\mathbb{P}(2.7 \leq X \leq 2.9)$.



$$P(X = x) = 0$$

$$P(a \leq X \leq b) = P(a < X < b)$$

$$P(2.7 < X \leq 2.9) =$$

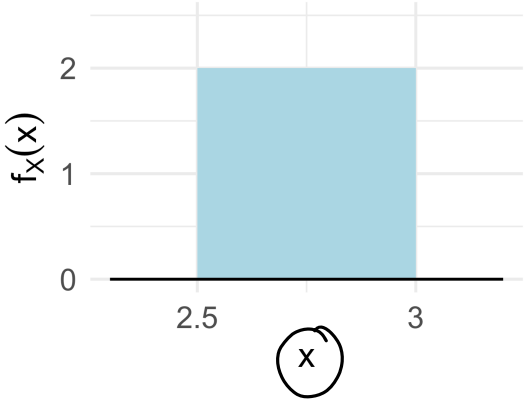
$$P(2.7 \leq X \leq 2.9) =$$

$$0.4$$

Example 1.4

Let $f_X(x) = 2$, for $2.5 \leq x \leq 3$.

4. Find $\mathbb{P}(X = 2.9)$.



$$P(X = 2.9) = \int_{2.9}^{2.9} 2 \, dx$$

$$= 2x \Big|_{2.9}^{2.9}$$

$$= 2(2.9) - 2(2.9)$$

$$= 0$$

$P(X \leq x)$ is the basis of the CDF

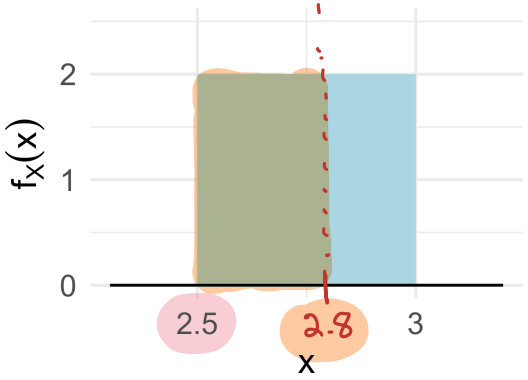
↳ p____() in R

↳ distributions

Example 1.5

Let $f_X(x) = 2$, for $2.5 \leq x \leq 3$.

5. Find $\mathbb{P}(X \leq 2.8)$. : implied prob $P(2.5 \leq X \leq 2.8)$



$$P(X \leq 2.8) = \int_{2.5}^{2.8} 2\,dx$$

$$= 2x \Big|_{x=2.5}^{x=2.8}$$

$$= 2(2.8 - 2.5)$$

$$= 0.6$$

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Use R to simulate known distributions (1/2)

normal
exponential

- We can use R to simulate continuous random variables and visualize their distributions
- For example, we can simulate a *uniform* distribution between 2.5 and 3: $f_X(x) = 2$ for $2.5 \leq x \leq 3$

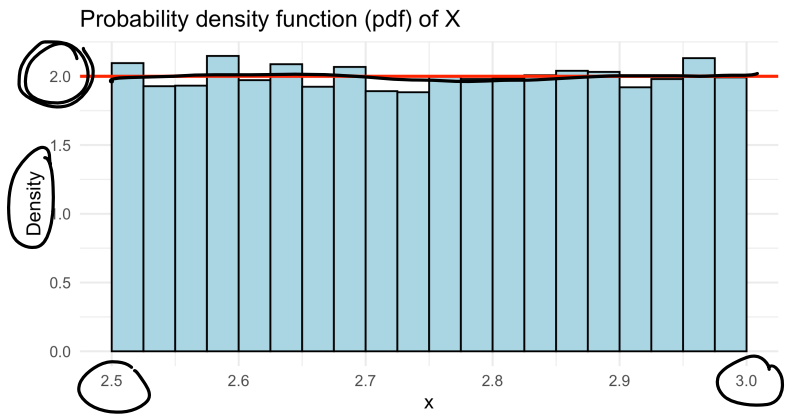
```
1 set.seed(8)
2 reps <- 10000
3 uniform_sims <- tibble(
4   x = runif(n = reps, min = 2.5,
5             max = 3)
6   )
7 head(uniform_sims)
```

↑ 100000

r: sample 10,000 x-values

```
# A tibble: 6 × 1
      x
  <dbl>
1  2.73
2  2.60
3  2.90
4  2.83
5  2.66
6  2.86
```

▶ Plot the simulated values and the pdf



$f_X(x) = 2$

$2.5 \leq x \leq 3$

Use R to simulate known distributions (2/2)

- We can use the simulated values to check our answers from Examples 1.2 to 1.5

```
uniform_sims |>
  summarise(
    prob_between = mean(x >= 2.7 & x <= 2.9),
    prob_equal   = mean(x == 2.9),
    prob_at_most = mean(x <= 2.8)
  )
```

P(2.7 ≤ X ≤ 2.9)

P(X = 2.9)

P(X ≤ 2.8)

```
# A tibble: 1 × 3
  prob_between prob_equal prob_at_most
         <dbl>      <dbl>        <dbl>
1        0.396          0        0.598
```

0.4

0.6

- Compare these to the answers we found with the integrals
- Does the simulation agree that $P(X = 2.9) = 0$?